Scalable Off-road Semantic Terrain Map Estimation using Uncertainty-aware LiDAR-Camera Fusion

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***Abstract*— Autonomous navigation in off-road conditions requires a precise semantic understanding of the surroundings, often represented in a bird's-eye view (BEV) for downstream tasks. Despite the success of end-to-end learning-based approaches in generating local semantic terrain maps from sensor data, their applicability in diverse off-road environments is hindered by limited 3D datasets and uncertain terrain characteristics. This paper proposes a scalable learning-based framework for generating dense semantic terrain maps in BEV. By fusing LiDAR and image data at multiple scales, our approach enhances the accuracy of the local semantic map estimated by an RGB image and a single LiDAR scan. Additionally, the network improves applicability across various off-road conditions without 3D annotations by leveraging uncertainty-aware, image-guided pseudo labels. Through extensive evaluation using diverse off-road driving datasets, we show that our method can improve both accuracy and adaptability in diverse off-road terrains, validating its efficacy in facilitating robust and reliable semantic terrain map estimation in varying off-road settings.**

I. INTRODUCTION

Autonomous navigation across complex and unstructured off-road terrains has become imperative for diverse robotic applications, including exploration, agriculture, search and rescue missions, and military operations. The success of off-road navigation depends on an accurate understanding of terrain characteristics and effective representation of this information for subsequent decision-making. Without prior environmental knowledge, off-road navigation systems must rely on real-time onboard sensor data to assess terrain attributes [1]. The terrains are often semantically classified and translated into Bird's Eye View (BEV) representations, facilitating integration into motion planning algorithms [2].

However, generating a reliable semantic terrain classification map poses significant challenges due to the distinctive characteristics of off-road environments, which diverge significantly from indoor or structured settings, such as uncertain terrain boundaries and diverse terrain types [3]. The high intra-class variation in terrain appearances introduces potential ambiguity in classification, and the diversity of terrain classes in off-road environments requires fine-grained labeling for comprehensive scene understanding [4], [5]. Moreover, the variability of terrain types across different scenes compromises the scalability of labeled datasets and pre-trained classification models [6]. Additionally, the complex geometry of off-road terrains complicates precise mapping in 3D spaces, as accurately estimating elevation and adopting flat-ground assumptions are challenging [7].

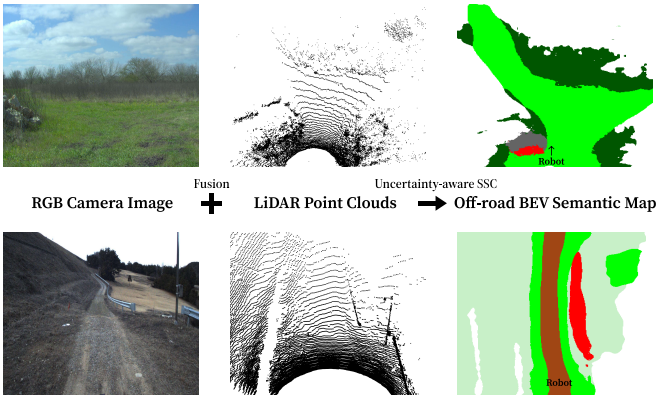


Fig. 1: Our method generates a BEV semantic terrain classification map from an RGB image and a single-sweep LiDAR point cloud. The classification accuracy is improved by combining complementary features extracted from the image and the LiDAR point cloud. Moreover, our approach utilizes uncertainty-aware pseudo-labels to perform semantic scene completion, resulting in a dependable, dense semantic map in diverse off-road settings.

These characteristics impose numerous restrictions on the reliable application of existing semantic terrain map estimation methods in various off-road environments. Conventional methods involve projecting pixel-wise segmentation into a BEV map using LiDAR or stereo cameras [1], resulting in sparse mapping due to the accumulation of point-wise predictions. Recent approaches leverage the concept of Semantic Scene Completion (SSC), employing learning-based inpainting to generate dense local classification maps [8], [9]. However, these approaches face limitations due to the inherently sparse and less semantically rich features of LiDAR measurements. Also, the scarcity of labeled 3D data for training limits their scalability to diverse off-road scenes. Some works explore learning-based models for terrain map estimation using only visual data, employing learning-based viewpoint transformations with 2D segmentation outputs [10]–[13]. Nevertheless, without explicit range measurements, these methods yield geometrically inaccurate and unreliable outcomes, posing potential risks during high-speed navigation [7]. Furthermore, their dependency on specific data distributions of labeled datasets makes extending their application to diverse environments challenging.

This paper presents a learning-based semantic terrain map estimation method to improve accuracy and scalability across diverse off-road scenarios, as shown in Fig. 1. Using a single LiDAR scan and an RGB image as input, the network generates dense and accurate terrain class maps in BEV by fusing information from multi-modal sensor measurements. To en-

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sure the scalability of the network training across various off-road scenes without relying on 3D annotations, pseudo-labels are generated through image-guided annotations. The reliability of the training using the pseudo-label is enhanced by uncertainty estimation, enabling the network to perform effectively in varied off-road environments. Comprehensive experiments have been conducted using our own off-road dataset and the publicly available RELLIS-3D dataset [14], verifying the scalability of our approach across diverse off-road environments. Moreover, our method shows improved accuracy compared to LiDAR-only and camera-only methods by incorporating the multi-modal feature fusion methodology for semantic terrain map estimation.

II. RELATED WORKS

A. Robotics Mapping in Bird's Eye View

Robotics mapping involves establishing a representation of a robot's surroundings using noisy measurements as it moves through an environment. The characteristics of these environments are assessed based on terrain features such as occupancy [15], traversability [16], or semantic class [17]. These representations are commonly converted into the BEV space due to their compatibility for integration with path planners in various robotic applications [16], [18].

For navigation in urban settings, camera-based methods widely employ viewpoint transformation learned by projecting pixel-wise features into BEV space [10]–[13]. Nevertheless, the practicality of these approaches in off-road settings faces challenges due to the complex geometry of off-road terrain. While other methods adopt range sensors such as LiDAR, the sparsity of LiDAR returns limits their effectiveness despite its high precision of geometric information [2], [16], [19]. Some methods perform semantic segmentation directly from raw sensor measurements, such as image or LiDAR, and then project the results onto BEV for the mapping [1]. However, the sparsity issues become more pronounced during high-speed navigation, where larger robot motion between LiDAR scans results in fewer depth measurements per unit area, compromising the reliability of the generated map.

Recent works propose learning-based approaches for predicting complete dense maps at a fixed size for off-road and unstructured environments to address these limitations [7], [9]. Specifically, they leverage the concept of Semantic Scene Completion (SSC) to generate a complete 3D scene from a single LiDAR scan [20], [21]. For instance, BEVNet [9] achieves dense and accurate off-road terrain semantic classification based on SSC. However, these methods often struggle to acquire accurate semantic predictions in off-road environments solely using LiDAR features. They also need 3D ground truth for consecutive scans, which poses a challenge in extending their model to varying off-road environments with diverse structures and class ontologies.

B. LiDAR-Image Fusion

While single-modal methods often face challenges in complex environments due to the inherent limitations of the input sensors, combining different modalities through sensor fusion has proven notable performance improvement in various applications [22]–[25]. LiDAR point clouds provide precise geometric information but only capture sparse data and lack texture information [26]. On the other hand, camera images can offer detailed and dense semantic information, while implicitly or explicitly inferred geometric information is prone to errors [24]. Hence, the combination of LiDAR and camera modalities is beneficial for performing terrain classification, as they can enhance each other's capabilities.

Input-level fusion methods employed a BEV or spherical projection to project image logits or features into LiDAR space to improve the performance of LiDAR-based networks [22], [23]. The feature-level fusion methods aim to enhance feature representation by combining and sharing information between the features of 2D and 3D backbones [24]. Notably, the 2DPASS [25] effectively leverages rich semantic information from images by transferring knowledge with multi-scale feature fusion across different modalities during the learning feature representations. Combining LiDAR and image modalities would be essential for enhancing semantic terrain map estimation in off-road scenarios, enabling the capturing of both intricate geometry and rich semantics.

III. METHODS

This section details our proposed learning method for generating a dense off-road semantic terrain map in BEV. First, a method for creating a reliable and scalable pseudo-ground truth is proposed. Then, the network structure is presented, which can generate a dense semantic terrain map in the robot's local frame from sensor measurements. LiDAR-image fusion is adopted to enhance prediction accuracy, and uncertainty-aware training is incorporated to increase the reliability of our method in various off-road settings.

A. Image-guided Pseudo Ground-truth Generation

While labeled 3D datasets can create BEV ground truth in constrained environments, their acquisition cost and limited applicability to specific sensor configurations and class definitions pose challenges in off-road conditions. To ensure reliability and scalability in various off-road settings, we adopt a pseudo-labeling approach for generating BEV ground truth. This strategy alleviates constraints associated with the scarcity and expense of 3D labels, enabling training across a wide range of off-road scenarios. The overview of our uncertainty-aware pseudo-labeling is depicted in Fig. 2

A pre-trained image segmentation network is utilized to produce the ground truth for semantic terrain classification. Image-based labels are beneficial in off-road scenes because they can accurately identify ambiguous boundaries and diverse classes, as RGB images are often characterized by rich details and textures. A dataset containing paired point clouds and RGB images can be easily obtained by navigating a robot equipped with LiDAR and a camera. The pseudo-labels of the BEV grid are then acquired by aggregating semantic segmentation predictions of the pre-trained model.

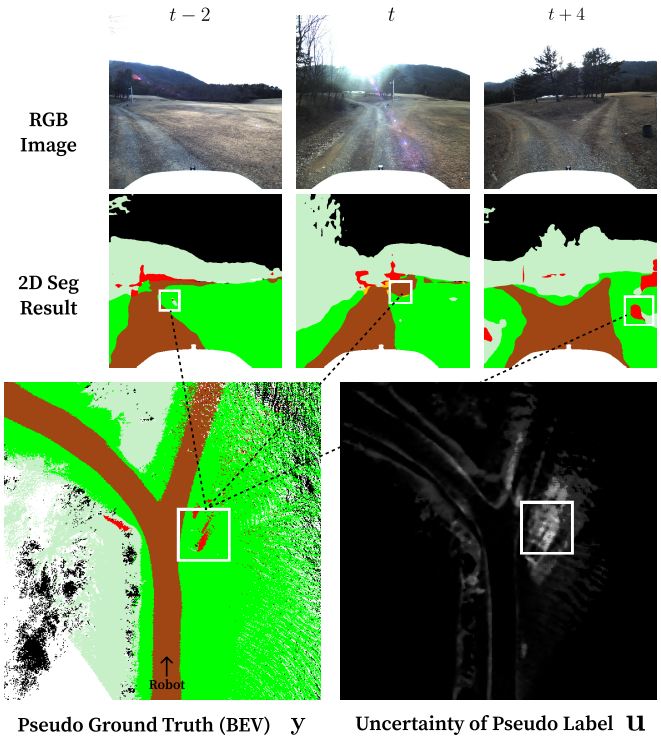


Fig. 2: Overview of image-guided pseudo-label generation. A pre-trained 2D image segmentation network derives semantic segmentation results from past and future images. These outcomes are aggregated using paired point clouds and then projected onto BEV grids to generate the pseudo-ground truth. Each grid determines a pseudo-label through the argmax operation, while its uncertainty is also quantified. Areas within the white box exhibit inconsistent semantic predictions across multiple timesteps, leading to higher uncertainties, depicted by brighter colors.

For each paired LiDAR and RGB image, the inference results of 2D semantic segmentation are aggregated from the past 50 and future 100 data instances. Given the image segmentation results at timestamp t', the prediction for each pixel is projected onto the BEV grid using the paired LiDAR point cloud. Given a ith LiDAR point $\mathbf{P}^i \in \mathbb{R}^3$ at timestamp t', the projection to pixel $\mathbf{I}^i$ in the image plane is determined based on camera parameters as follows:

$$\mathbf{I}^i = \mathbf{K} \cdot \mathbf{T} \cdot \mathbf{P}^i, \tag{1}$$

where $\mathbf{K} \in \mathbb{R}^{3\times4}$ and $\mathbf{T} \in \mathbb{R}^{4\times4}$ are the camera intrinsic and extrinsic matrices, respectively. The point is pseudo-labeled with a class probability $\mathbf{S}^i \in \mathbb{R}^K$, where K is the total number of classes.

To enhance the reliability of pseudo-labels generated from the pixel-wise semantic probability $\mathbf{p}^i \in \mathbb{R}^K$, the uncertainty of the semantic probability is incorporated into the pseudo-labeling. The normalized entropy function is defined as follows to calculate the uncertainty of semantic segmentation:

$$E(\mathbf{p}^i) = -\frac{1}{\log K} \sum_k^K \mathbf{p}_k^i \log(\mathbf{p}_k^i + \epsilon), \tag{2}$$

where ϵ is the coefficient for avoiding numerical error. Then, the confidence of a semantic prediction at i^{th} pixel, denoted as C^i, is calculated as :

$$C^i = 1 - E(\mathbf{p}^i). \tag{3}$$

Each point is labeled with a soft class probability weighted by its certainty, imposing greater emphasis on confident predictions over uncertain ones as follows:

$$\mathbf{S}^i = C^i\, \mathbf{p}^i. \tag{4}$$

The labeled points are then transformed into the reference robot frame at timestamp t as $\mathbf{P}^i_{t'\to t}$, based on the robot pose recovered by SLAM or odometry [27]. Multiple predictions are merged into a reference frame to produce dense BEV labels, and points from multiple timestamps are rasterized into BEV grid cells based on their x and y positions. A label for each grid G^j is determined from the labeled points assigned to the grid, while some grids without any assigned points are labeled as an unknown class. To avoid the aliasing of moving objects during aggregation, only points that belong to static object classes are aggregated, while points that belong to moving objects are aggregated only if they are from 3 sequential scans from the current scan.

Soft class probabilities of points assigned in the same grid are summed for each class to calculate the grid class score, $\mathbf{c}^j \in \mathbb{R}^K$, where j is the index of the BEV grid. Then, the pseudo-label of a grid, $\mathbf{y}^j$, is determined through a majority vote of class predictions:

$$\mathbf{c}_k^j = \sum\nolimits_{i \text{ s.t. } \mathbf{P}^i_{t'\to t} \in G^j} \mathbf{S}_k^i, \tag{5}$$

$$\mathbf{y}^j = \arg\max_k \ \mathbf{c}_k^j. \tag{6}$$

Adopting multiple segmentation inference outcomes through majority voting introduces ensemble-like effects, effectively addressing potential inaccuracies in 2D segmentation results.

Additionally, the uncertainty of the pseudo-label of grid j, denoted as $\mathbf{u}^j \in [0, 1]$, is calculated to measure the reliability of the pseudo-label by measuring the consistency of semantic predictions for a grid over multiple timestamps:

$$\mathbf{u}^j = E(\mathbf{c}^j/\alpha^j), \tag{7}$$

where $\alpha^j = \sum_k^K \mathbf{c}_k^j$ is the normalizing factor and $E(\cdot)$ is the entropy function defined in (2). These uncertainty estimates can be leveraged during the network training to enhance the robustness of semantic terrain map estimation that is trained with noisy pseudo labels [28].

B. BEV Semantic Fusion Network

Our network is trained to generate a dense top-view semantic classification map of local surroundings, utilizing a sparse LiDAR point cloud and a paired RGB camera image. The overview pipeline of our network is depicted in Fig. 3.

LiDAR and image features are extracted from separate networks in 3D voxel spaces. The input LiDAR point cloud undergoes discretization into a (H, W, D) voxel grid with a resolution of $0.1m \times 0.1m \times 0.2m$. In each voxel, a point is encoded as a 3-dimensional feature, comprising the offset from the voxel center $(\Delta x, \Delta y, \Delta z)$. Utilizing a simplified PointNet [29] architecture that includes a linear layer, BatchNorm, and ReLU, each voxel of size $(N, 3)$ is transformed into sparse LiDAR voxel features of size

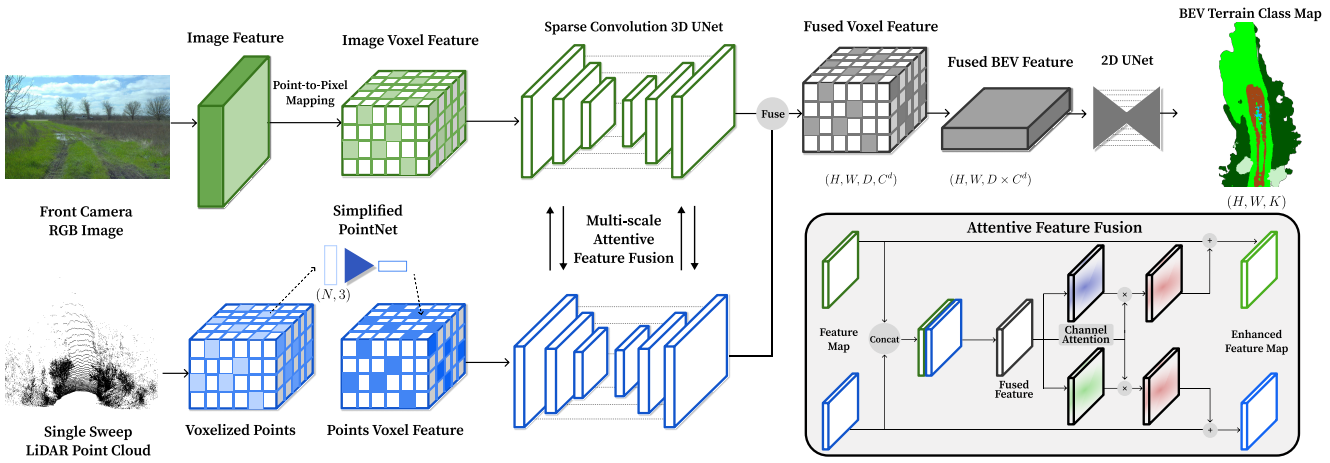


Fig. 3: High-level architecture of the proposed network. The network takes input from a single-sweep LiDAR point cloud and an RGB image captured by the front camera, producing a dense semantic terrain classification map in BEV. Extracted features from the image and point cloud, obtained through distinct encoders, are fused using Multi-scale Attentive Feature Fusion, integrated into each level of a 3D UNet for each modality. Subsequently, these fused features are passed to a 2D UNet to generate the dense semantic terrain classification map of local surroundings in BEV.

C, where N is the maximum number of points per voxel. Simultaneously, image features are extracted from a pre-trained image backbone and similarly converted into sparse image voxel features. The LiDAR points are projected onto the image plane via point-to-pixel mapping based on (1), and the pixel-wise image features are propagated to the corresponding voxel.

The input features from RGB and LiDAR are independently passed through separate 3D U-Net composed of sparse convolution layers to extract features for each modality. The 3D U-Net architecture employs a multi-level encoder-decoder structure, where each decoder layer is connected to the encoder of the same level through a residual skip connection. After each level of the 3D U-Net, a multimodal fusion block is integrated to promote feature fusion between LiDAR and image features. This facilitates the effective blending of rich semantic information from RGB with the geometric details derived from LiDAR points, ultimately leading to the generation of a precise semantic terrain classification map.

For fusing image and LiDAR features at each level, attentive feature fusion is employed to complement the features from each modality effectively, as shown in Fig. 3. Image and LiDAR features are concatenated channel-wise, followed by sparse convolution to generate a fused feature map. Channel attention is applied to the fused feature for each modality, emphasizing features that can strengthen the information from complementary sensors. The attended fused features are then added back to the original features of each modality, enabling a concentration on the more crucial information and distilling features from different modalities.

From the fused voxel features, a dense BEV terrain classification map is produced using a 2D convolution network. The voxel feature map, with dimensions (H, W, D, C^d) undergoes compression to yield a BEV feature map of dimensions $(H, W, C^d \times D)$ with empty grids initialized to zero. A U-Net-structured 2D convolution network is employed to generate a dense semantic map. This network progressively reduces the spatial size of features, capturing higher-level semantic information, while the decoder part of the network upsamples feature maps to recover spatial information. Through this process, every grid feature is interpolated from sparse features to produce a dense terrain classification map of size (H, W, K) after a set of 1×1 convolutions in the segmentation head, which outputs logits for the classes.

C. Uncertainty-aware Semantic Terrain Map Estimation

The model is optimized using the uncertainty-aware pseudo-label produced in Section III-A with uncertainty-weighted cross-entropy loss. To impose a lower weight on the ground truth with high uncertainty stemming from inconsistent pseudo-labels, the cross-entropy loss is calculated as follows:

$$\mathcal{L}_{\text{cls}} = -\sum_{j} \frac{\sum_k \mathbf{Y}_k^j \log \hat{\mathbf{Y}}_k^j}{1 + \mathbf{u}^j / \tau}, \tag{8}$$

where $\hat{\mathbf{Y}}^j \in \mathbb{R}^K$ is softmax probability for grid j produced by the network, $\mathbf{Y}^j \in \mathbb{R}^K$ is one-hot vector representation of the pseudo-label $\mathbf{y}^j$, and τ is the coefficient for controlling smoothness. By assigning weights inversely proportional to uncertainty defined by (7) in the cross-entropy loss term, the contribution of certain labels can be enhanced during optimization while that of uncertain labels is minimized. By employing this strategy, the network can be effectively trained in diverse off-road environments without the need for manual annotations for each individual point cloud to create BEV semantic ground truth, as this approach helps alleviate the adverse effects of uncertainty in pseudo-labeling.

IV. EXPERIMENTS

In this section, we validate the effectiveness of our method in enhancing semantic terrain map generation performance in off-road environments. Firstly, we introduce the dataset used for experiments, including publicly available datasets and our off-road datasets. Through a quantitative and qualitative analysis, the results obtained from our method are compared with

TABLE I: Quantitative results on the RELLIS-3D dataset [14] and our off-road driving dataset.

	Method	Acc [%]	mIoU [%]	void	dynamic	static	road	grass	dirt	puddle	rubble	tree	bush
RELLIS 3D	*PyrOccNet* [10]	30.0	18.6	89.5	0.0	4.1	22.8	37.8	3.7	4.7	14.9	4.5	8.5
	TIM [11]	30.5	17.6	88.8	0.0	1.2	9.6	36.6	**6.1**	3.2	13.5	7.2	9.4
	BEVNet [9]	50.7	31.6	91.6	3.6	11.1	40.1	**54.9**	0.0	0.3	35.0	54.3	25.0
	Ours	**52.9**	**36.1**	**92.1**	**4.8**	**23.4**	**50.9**	53.7	5.8	**11.6**	**37.8**	**54.6**	**26.1**

	Method	Acc [%]	mIoU [%]	void	paved	unpaved	grass	vegetation	puddle	structure	static	dynamic
Our Dataset	*PyrOccNet* [10]	39.9	26.1	59.5	23.5	50.8	45.5	39.4	0.0	10.8	5.3	0.0
	TIM [11]	40.9	25.6	43.3	**27.1**	51.7	44.3	42.0	0.0	14.2	**7.4**	0.0
	BEVNet [9]	39.1	26.2	57.5	5.6	53.5	51.0	52.1	0.0	15.1	0.1	0.8
	Ours	**47.1**	**32.6**	**61.4**	23.3	**60.9**	**58.7**	**56.3**	0.0	**20.9**	6.8	**5.2**

those of other existing approaches. This evaluation focuses on assessing the effectiveness of LiDAR-image fusion for terrain map estimation and confirming the scalability of our method through uncertainty-aware optimization.

A. Dataset

1) RELLIS-3D: We present our experimental results utilizing the publicly available off-road dataset, RELLIS-3D [14]. This dataset contains RGB camera images and LiDAR point clouds, accompanied by point-wise semantic annotations and accurate robot poses recovered by SLAM [30]. We utilized sequences 0, 1, 2, and 4 for training and sequence 3 for evaluation. To address the class imbalance, some similar minor classes are grouped into a single category, resulting in a total of 10 classes, as shown in Tab. I.

2) Our Off-road Driving Dataset: In addition to the RELLIS-3D dataset, we collected a supplementary driving dataset encompassing much more diverse off-road terrains using our platform, which is equipped with an OS1-128 LiDAR, front-view RGB camera, and IMU [16]. This dataset comprises approximately $18,000$ images captured under various conditions, including different seasons, lighting scenarios, and terrain types. For training and evaluation purposes, RGB images are manually annotated by experts. The dataset follows a $5:1$ split ratio, resulting in $14,486$ samples for training and $3,711$ samples for validation.

B. Experimental Setup

1) Implementation Details: For all experiments, the input point cloud is cropped at $[(0, 51.2), (-25.6, 25.6), (-3, 5)]$ meters along the x, y, z axes, and a voxel grid size of $0.1m \times 0.1m \times 0.2m$ is used, resulting in dimensions $(H, W, D) = (512, 512, 40)$. The maximum number of points per voxel is set to $N = 10$, and the channels of voxel features are set to $C = 64$. For the 2D segmentation backbone, we adopt DeepLabV3 with ResNet50. The 3D U-Net sparse convolution network has encoder and decoder parts, each with four layers, and an attentive feature fusion block is attached to every layer. The LiDAR encoder has one *SparseConv* and two *SubMConv* for each layer, while the image encoder has only one *SubMConv* for each layer. This design is because the pre-trained 2D image segmentation model has already extracted rich features for images.

We train our model using the Adam optimizer, with a learning rate of $3e^{-4}$ and a batch size of 8 for 30 epochs. During training, point clouds are randomly augmented with vertical flipping, translations in the x and y axes within the range of (-5, 5) meters, and rotations around the z-axis within the range of $(-\frac{\pi}{4}, \frac{\pi}{4})$ radians.

2) Ground-truth Generation: To establish the top-view ground truth for training, we accumulate point clouds from consecutive LiDAR scans, aligning them with the LiDAR coordinate frame of the current scan, as explained in Section III-A. Similarly, for the evaluation dataset, the top-view semantic map ground truth is determined using a similar approach, while we leverage 3D annotation for ground truth generation for RELLIS-3D as the dataset provides it [14].

3) Comparison Methods: Our method is compared to relevant approaches for generating the local semantic terrain classification map in BEV. First, image-only methods that conduct view transformation to convert monocular images to BEV semantic maps are utilized. The Pyramid Occupancy Network (*PyrOccNet* [10]) employs a multiscale convolution network architecture to produce dense map representations directly from monocular images in BEV. Translating Images into Maps (*TIM* [11]) addresses generating BEV maps from images by solving sequences-to-sequences translation problems through an attention-based architecture. We also compare our method with a LiDAR-only approach that generates a dense semantic terrain classification map in off-road conditions, *BEVNet* [9]. Note that all models are evaluated solely with a LiDAR point cloud or RGB image of the current timestamp without using methods for temporal aggregation, such as recurrent neural networks, to objectively focus our evaluation on local BEV terrain map estimation quality from sensor measurements.

4) Evaluation Metrics: Intersection over Union (IoU) is employed to assess the performance of semantic terrain classification in BEV. The IoU for class k is calculated as:

$$\mathrm{IoU}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k}, \tag{9}$$

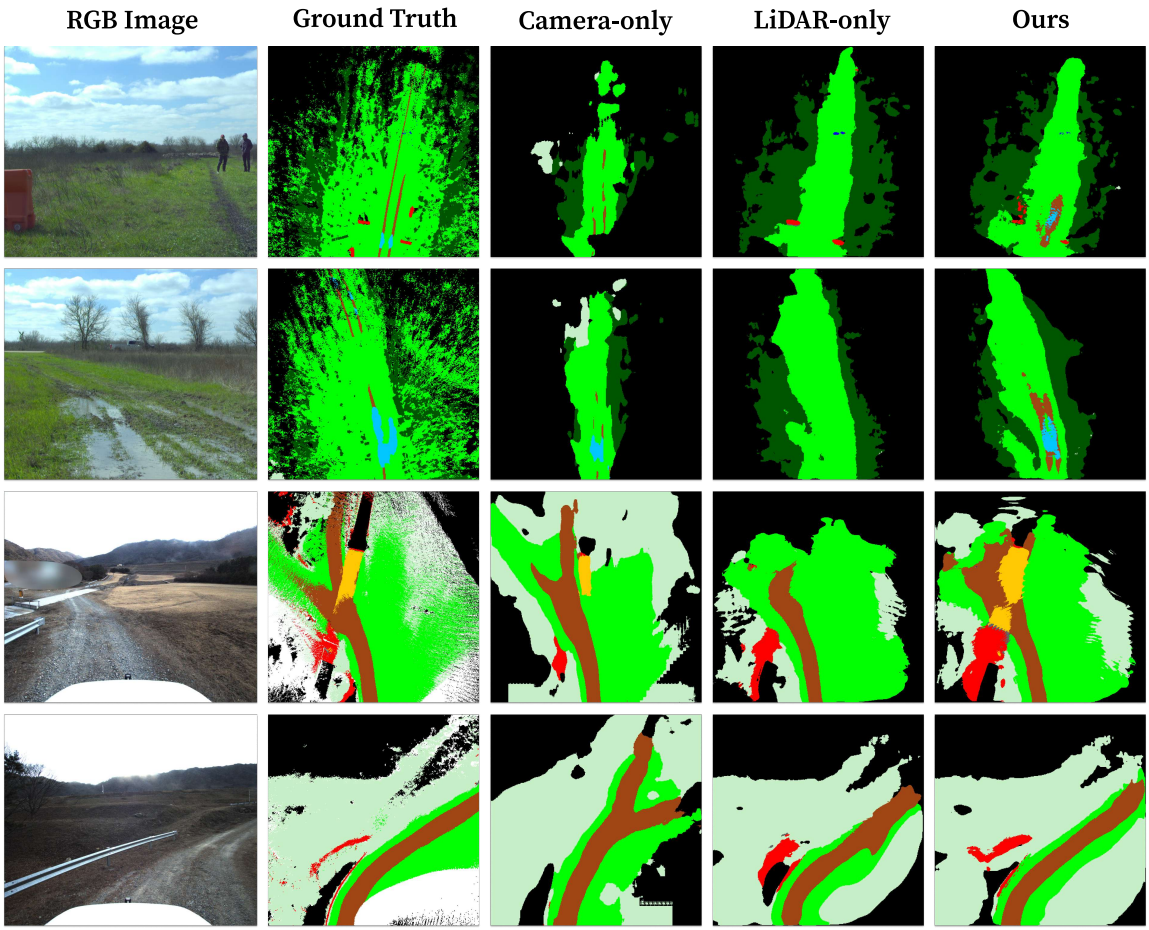


Fig. 4: Compared to other methods, *Ours* successfully predicted a semantic terrain map with the LiDAR-image fusion. For instance, our method accurately identifies puddles and dirt in BEV maps while preserving the geometrical structure of the scenes. The camera-only method excels at extracting semantic information from terrain but fails to represent a map with accurate geometrical structure of complex off-road scenes. On the other hand, LiDAR-only methods accurately represent geometrical information but are vulnerable to the semantic classification of terrain.

where $\mathrm{TP}_k, \mathrm{FP}_k$, and FN_k, represent the number of true positive, false positive, and false negative predictions in grids, respectively. To evaluate the overall performance, the mean IoU (mIoU) and the accuracy are computed as:

$$\mathrm{mIoU} = \frac{1}{K}\sum_{k=1}^{K} \mathrm{IoU}_k, \tag{10}$$

where K is the total number of classes. For a comprehensive evaluation of our semantic terrain classification model, we additionally present overall prediction accuracy computed as:

$$\mathrm{Accuracy} = \frac{\sum_{k=1}^{K} \mathrm{TP}_k}{\sum_{k=1}^{K} (\mathrm{TP}_k + \mathrm{FP}_k)}. \tag{11}$$

Please note that unlabeled grids, which consist of points not labeled through image-guided labeling, are excluded from the evaluation with our off-road driving dataset.

C. Experimental Results

The quantitative results for the RELLIS-3D dataset and our off-road dataset are presented in Tab. I. In both datasets, our approach outperforms other methods specifically designed for structured conditions or that rely on a single modality. Leveraging LiDAR-image fusion, our method outperforms baseline methods in terms of mIoU and Accuracy.

Our fusion approach demonstrates improved IoU scores, particularly for challenging classes like dirt and puddles, where single-modal methods struggle. Image-only view-transformation methods perform poorly in off-road environments, especially for grass, road, tree, and bush classes containing intricate geometrical structures. LiDAR-based methods, on the other hand, exhibit more robust performance when predicting terrains with complex shapes. However, the semantic prediction accuracy of the LiDAR-only methods is inferior to ours, which improves reliability by utilizing uncertainty-aware optimization and feature-level fusion. Specifically, LiDAR-only methods exhibit inferior performance in predicting classes requiring a higher-level understanding of textures, such as dirt and puddles.

Qualitative results for both datasets are presented in Fig. 4. The first and second rows depict the outcomes for RELLIS-3D, while the subsequent rows show the results for our off-road dataset. Our method effectively produces local semantic terrain maps with precise structures by incorporating LiDAR features. Furthermore, our approach exhibits enhanced precision in estimating semantic terrain classes with classes that demand a nuanced understanding of textures.

D. Ablation Study

We present ablation studies to examine the validity of each component of our methodology. Tab. II provides quantitative validation of the effectiveness of incorporating the LiDAR-image fusion component and uncertainty-aware optimization.

TABLE II: Results of the ablation studies.

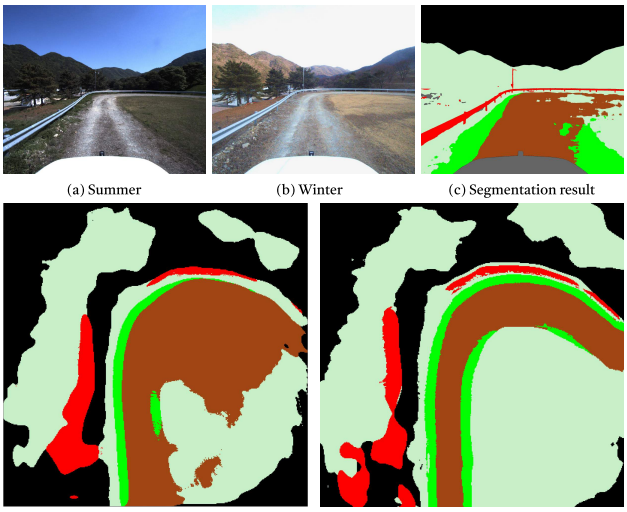
Module		RELLIS-3D		Our Dataset	
Fusion	**Uncertainty**	mIoU	Acc	mIoU	Acc
✗	✗	29.2	48.1	27.0	42.5
✗	✓	28.2	48.9	26.5	41.4
✓	✗	35.2	52.5	31.2	46.0
✓	✓ *(hard)*	35.8	51.4	31.6	45.9
✓	✓ *(soft)*	**36.1**	**52.9**	**32.6**	**47.1**

The models without the fusion component are trained only using LiDAR features. Notably, incorporating multi-modal fusion significantly improves performance in both datasets, indicating strengthened features through fusion. While solely using uncertainty-weighted cross-entropy does not significantly enhance performance, it improves results when combined with the fusion methodology. This suggests that uncertainty-aware optimization effectively mitigates training uncertainty arising from imprecise labels and addresses the uncertainty of image features in pre-trained models. Although accumulating one-hot hard predictions in (5) instead of soft probability as in (4) for uncertainty-aware optimization (*hard*) shows performance improvement, the enhancement is even greater when using the soft class probabilities. Uncertainty-aware optimization with soft probabilities enables a more detailed consideration of uncertainty (*soft*), incorporating both temporal consistencies through multiple prediction accumulations and more precise delineation of the uncertainty in image segmentation.

TABLE III: Results of the ablation for LiDAR-image fusion.

Fusion Method	RELLIS-3D		Our Dataset	
	mIoU	Acc	mIoU	Acc
Early fusion	29.7	47.7	26.1	40.8
Feature fusion (w.o. attention)	33.8	51.7	28.9	40.3
Feature fusion (w.o. multi-scale)	34.4	51.5	29.0	41.5
Feature fusion (Ours)	**35.2**	**52.5**	**31.2**	**46.0**

We then quantitatively evaluate the efficacy of our LiDAR-image fusion component, which fuses features of each modality on multiple scales with an attention mechanism. For comparison, we train the network with different fusion strategies. First, *Early Fusion* simply concatenates image and LiDAR features before 3D U-Net and utilizes a single sparse convolution 3D U-Net. Then, we train the network with several different *Feature Fusion* strategies: *w.o. attention*, which indicates that no channel attention is applied during the fusion, and *w.o. multi-scale*, which involves the attentive feature fusion only in the last layer of the 3D U-Net. To objectively assess the contribution of fusion methodologies, these models are compared with the model learned using our approach without uncertainty-aware loss (*Ours*).



(a) Summer (b) Winter (c) Segmentation result

(d) Semantic map estimation w/o uncertainty (e) Semantic map estimation with uncertainty

Fig. 5: (a), (b) Off-road environments exhibit significant changes in visual appearances depending on various factors, such as the season and lighting conditions. (c) The segmentation model pre-trained on the summer subset shows uncertain predictions when applied to the winter subset. (d), (e) By incorporating the uncertainty-aware pseudo-label, our method can robustly estimate semantic terrain maps despite the domain shift.

Experimental results in Tab. III demonstrate the effectiveness of our network design on both datasets. While early fusion methods show improved results over the LiDAR-only model, they perform less effectively than feature-level fusion methods. This implies that more than simply decorating point features with image features is required to ensure the supplementation of the features effectively. Incorporating attention during fusion enhances performance, indicating the benefits of leveraging the channel attention mechanism to share appropriate information from each complementary feature. Finally, multi-scale fusion further improves results, underscoring the efficacy of fusing features at multiple resolutions, which aligns with prior research [25].

E. Scalability in Off-road Environments

Due to the complexity of off-road environments, obtaining reliable pre-trained models for every setting is challenging, often leading to the presence of noisy and uncertain predictions. Therefore, we verify the scalability of our method, particularly in scenarios involving variability in visual appearances. To illustrate this scenario, we test our method's scalability under domain shift by using a 2D segmentation model trained on the summer subset of our off-road dataset to generate BEV pseudo-labels in the winter subset.

While being the same scene, the variability in visual appearances between the two images is evident, as shown in Fig. 5. Our model trained with noisy pseudo-labels generated from these unreliable predictions without the uncertainty-aware optimization (8) misclassifies a significant portion of vegetation as dirt, as seen in Fig. 5d. On the other hand, the model trained with uncertainty-aware optimization with (8) accurately represents most misclassified areas as shown in Fig. 5e. This highlights the robustness of network training even with noisy pseudo-labels, suggesting the increased applicability across diverse off-road environments.

V. CONCLUSION

This paper introduces a method for estimating BEV semantic terrain classification maps in off-road environments, leveraging LiDAR-image fusion to enhance mapping accuracy and uncertainty-aware pseudo-labeling to improve scalability. By utilizing a single LiDAR scan and a frontal RGB image as input, the network employs attentive feature fusion at multiple scales to extract richer terrain features. Additionally, uncertainty-aware pseudo-labels are generated through image-guided annotations, enabling the network to learn effectively in diverse off-road scenarios without manual 3D annotations. The training's reliability with pseudo-labels is enhanced by uncertainty-aware loss, assigning lower weights to grids with high uncertainties during training. Evaluation across varied off-road driving datasets validates the method's efficacy for semantic terrain map estimation.

Limitations and Future Works Although this method effectively produces precise semantic classification maps of off-road environments, it tends to generate overconfident predictions, a common issue in learning-based map representations. During interpolation from limited sensor measurements to infer semantic occupancy of grid cells, regions with occlusions may lead to overconfident predictions due to high uncertainties regarding terrain characteristics. Moreover, there is still untapped potential in leveraging images for BEV map generation. The method could benefit from incorporating view-transformation techniques, depth estimation, or recent advancements in transformer architecture.

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